Chat & Group Management Web App

Web App Design & Technical Architecture

A modern real-time communication platform where users can chat, create/join groups and channels, and administrators can manage members and administrative roles.

1. Core Features

Feature	Description
User Chat	Private one-to-one conversations with text messages, timestamps and online status.
Groups	Users can participate in groups. Admins can create groups and manage members.
Channels	Create channels for announcements, communities or topic-based communication.
Add Users	Group admins can search/select users and add or invite them.
Change Admin	The current admin can transfer administrative privileges to another member.
Remove Users	Admins can remove members according to group permissions.
Authentication	Secure registration, login and account management.
Responsive UI	Works across desktop, tablet and mobile screen sizes.

2. System Architecture

Users
↓
Frontend — Chat interface, groups, channels, profiles and admin screens
↓ API requests
Backend on Render — Authentication checks, messaging logic, group management and admin permissions
↓
Supabase — Authentication + PostgreSQL database

Deployment: Frontend → Netlify Backend → Render Database/Auth → Supabase

3. Technology Stack

Technology	Purpose
Frontend	Build the user interface and interactive chat experience.
Lato	Primary web font for a clean, readable visual style.
Netlify	Deploy and host the frontend application.
Render	Deploy and host the backend/API.

Technology	Purpose
Supabase Auth	User registration, login and authentication.
Supabase PostgreSQL	Store users, groups, memberships, messages and channels.
Backend API	Securely handle group administration, permissions and database operations.

4. Main Screens

- Login / Sign Up: Email, password, account creation and password recovery.
- Chat Dashboard: Search, recent chats, groups, channels and notifications.
- Private Chat: Conversation area, message composer, timestamps and online status.
- Group Page: Group information, member list, messages and settings.
- Channel Page: Channel information, posts/messages and member access.
- Admin Panel: Add users, remove users, change admin and manage group settings.

5. Database Structure

Table	Important Fields
users	id, name, email, profile_image, created_at
groups	id, name, description, created_by, created_at
group_members	id, group_id, user_id, role (admin/member)
messages	id, sender_id, receiver_id, group_id, message, created_at
channels	id, name, description, created_by, created_at

6. Admin Workflow

Create Group → Add Members → Assign Admin → Manage Members → Transfer Admin → Update Group Settings.

Only authorized administrators should be allowed to perform sensitive actions such as adding/removing members or transferring the admin role. These permissions should be enforced in the backend, not only hidden in the frontend.

7. UI / Design Direction

- Clean modern communication-app interface.
- Lato as the main web font.
- White/light-gray surfaces with a blue accent color.
- Rounded message bubbles and cards.
- Desktop: three-column layout — sidebar, chat area, members/admin panel.
- Mobile: collapsible sidebar and members panel for a compact chat experience.
- Clear role badges such as Admin and Member.

Project summary: A scalable chat and community platform with a Netlify frontend, Render backend, and Supabase authentication/database.